



Simulation of a Hypersonic Busemann Inlet at Off Design Mach Number and Angle of Attack

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In this study, three-dimensional simulations of a hypersonic streamtraced Busemann inlet at sub design Mach number, and angle of attack are performed using a positivity-preserving and entropy bounded discontinuous Galerkin formulation. The inlet is designed for operation at Mach 5.5 and the simulations are run at Mach 3.7 at angles of attack of 0° and $\pm 5^\circ$. The simulations are validated against available experimental pressure sensitive paint measurements and schlieren imaging. The shock structure and pressure distributions are examined at each angle of attack to identify the change in flow features. Additionally, downstream planes are extracted to determine the mass capture and mass flow lost through the bleed slot under each operating condition. When the inlet is pitched up there is an increase in the pressure rise across the inlet, decrease in the Mach number, and increase in the total mass flow captured. To examine the flowfield further the total pressure recovery and turbulent kinetic energy are examined at several planes along the inlet. It is shown that at each angle of attack, the flowfield exhibits distinct characteristics. At angles of attack of 0° and -5° , regions of low total pressure recovery and high turbulent kinetic energy are primarily concentrated along the upper half of the inlet. In contrast, at an angle of attack of $+5^\circ$, areas of high turbulent kinetic energy and low total pressure are distributed across the entire inlet cross-section.

I. Introduction

IMPROVED understanding of the flow-field dynamics within high-speed airbreathing propulsion vehicles is necessary for further developing hypersonic flight vehicles. Notably the scramjet is an attractive option for enabling air-breathing hypersonic because it relies on aerodynamic compression and heating due to intrinsic flow features from shock waves, instead of relying on rotating mechanical components (e.g. compressor) [1]. Although the thermodynamic cycle of a scramjet engine is relatively simple and well understood, designing an engine to operate across a large range of flight conditions and vehicle attitudes remains a significant challenge. The scramjet consists of four main components: the inlet, isolator, combustor, and nozzle. The inlet plays a critical role in vehicle operation, as they are responsible for supplying the combustor with supersonic airflow at an acceptable temperature, pressure, and mass flow rate [2, 3]. Common inlet designs, however, are typically optimized for specific flow speeds and conditions, resulting in losses under off-design conditions [4].

The most common inlet type explored are ramp based inlets. These inlets are typically two-dimensional and rely on multiple ramps to compress the flow [5–8]. While there are advantages to these inlets due to their simplicity, such as airframe integration, there also shortcomings such as low total pressure recovery [6]. Additionally, several three-dimensional inlet designs using streamline tracing methods (e.g. Busemann Inlets) have become more intensively researched in recent years [4, 9, 10]. These inlets are designed by obtaining the solution to the Taylor-Macoll equations for a selected Mach number and contraction ratio [11]. These inlets are advantageous as they rely on isentropic compression,

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which allows them to have superior inlet efficiency due to the improved total pressure recovery ratio [11–13]. However, challenges with these inlet configurations arise as they are designed for a single Mach number and with zero angle of attack (α). This results in significant unknown behavior for airbreathing propulsion vehicles as it is necessary to maintain consistent and stable operation across different flight and attitude conditions.

Several research efforts have looked into the on design performance of these streamline traced inlets [11, 14–17]. However, fewer studies have investigated off-design performance at varying Mach number or angles-of-attack. An experimental study by Di Cristina et al. used planar laser Rayleigh scattering to examine the internal flow characteristics of a stream traced inlet at on and off-design Mach number conditions [18]. The results indicated that at sub-design Mach numbers the general shock and flow features remained similar and that altering the Reynolds number resulted in thickened boundary layer behavior. Additionally, an experimental study performed by Schram et al. [10] analyzed a streamlined traced inlet at off design conditions and changes in angle of attack. It was shown that the mean pressure distribution increased under positive angle of attack orientation and decreased under negative angle of attack.

Additionally, several numerical studies have been performed examining inlet performance. Mcready et al. simulated a Busemann inlet at sub-design Mach number using multiple solvers and turbulence modeling approaches [19]. The simulations were able to capture the large region of flow separation on the inlet forebody and demonstrated that a Menter-type RANS model was able to capture the flow physics adequately. Jung and Im performed Reynolds Averaged Navier Stokes (RANS) simulations of a scoop inlet at sub-design to examine the relationship between Mach number and the angle of attack at which the inlet unstarts [7]. Specifically, they found that operation at lower Mach numbers resulted in unstart at progressively lower angles of attack. Bisek performed unsteady Large Eddy Simulations on the HIFiRE-6 vehicle, which included a streamtraced inlet and examined the effects of off-design operation [20, 21]. It was shown that the off design operation changed the captured mass flow and the pressure distribution along the inlet.

The current work examines three-dimensional simulations of the North Carolina State University hypersonic streamtraced Busemann inlet from Schram et al. [10]. Implicit Large Eddy Simulations (ILES) are performed using an extended version of the JENRE[®] Multiphysics Framework with the discontinuous Galerkin scheme presented by Ching et al. [22]. The inlet is run at a sub design Mach number of 3.7 at $\alpha = 0^\circ$ and $\alpha \pm 5^\circ$. The simulations are validated against pressure sensitive paint measurements and schlieren imaging from the corresponding experiment. Results are presented to compare off-design operation using flowfield characteristics such as mass flow rate, compression ratio, Mach number, and total pressure recovery. Additionally, results are provided to illustrate variations in local turbulent characteristics under different angle of attack conditions, revealing how the flowfield is altered as a result of changes in angle of attack.

II. Numerical Methodology

In this study, a fully conservative formulation of the multi-component, Navier Stokes equations [23] are solved, which are defined as

$$\frac{\partial (\rho e_t)}{\partial t} + \nabla \cdot ((\rho e_t + p) \mathbf{v}) = \nabla \cdot \left(\lambda \nabla T - \sum_{i=1}^{N_s} W_i C_i h_i \mathbf{V}_i + \boldsymbol{\tau} \cdot \mathbf{v} \right), \quad (1)$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p \mathbb{I}) = \nabla \cdot \boldsymbol{\tau}, \quad (2)$$

where ρ is density, $\mathbf{v}$ is velocity, p is pressure, $\mathbb{I}$ is an identity matrix, $\boldsymbol{\tau}$ is the deviatoric stress tensor as defined in Eq. (2.17) in [23], e_t is total energy, λ is conductivity, T is temperature, N_s is the number of species, W_i is molecular weight, C_i is concentration, h_i is enthalpy, $\mathbf{V}_i$ is diffusion velocity, ω_i is the production source term, and subscript i indicates species i . The production source term is calculated from the progress reaction rates for any number of reactions and reaction types. Pressure is determined using the equation of state,

$$p = R^\circ T \sum_{i=1}^{N_s} C_i, \quad (3)$$

where R° is the universal gas constant. The species diffusion velocity is defined by

$$\mathbf{V}_i = \frac{D_i}{C_i} \nabla C_i - \frac{D_i}{\rho} \nabla \rho, \quad (4)$$

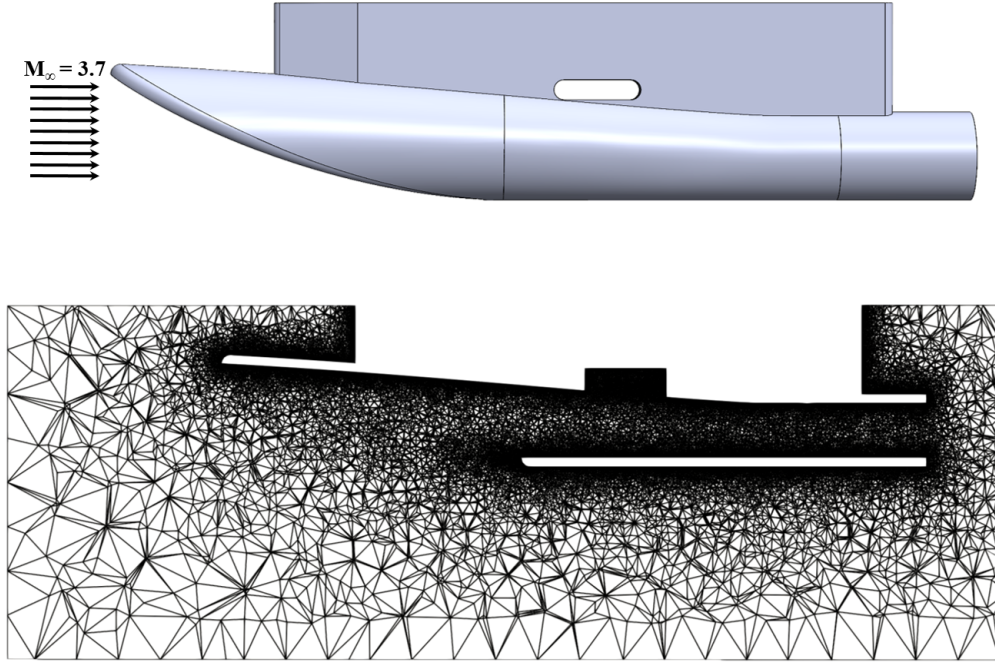


Fig. 1 Geometry and mesh for the 0° angle of attack

where D_i is the mass-averaged diffusion coefficient.

These equations are solved using the high-order discontinuous Galerkin method, in which the element-local solution is approximated using polynomials of arbitrary order, denoted p . We employ the HLLC inviscid flux function and the BR2 viscous flux function. Hartmann-type artificial viscosity is used to stabilize shocks, and a linear-scaling limiter is used to guarantee that species concentrations are nonnegative, pressure and temperature are positive, and entropy is bounded. More information on the discretization and stabilization strategies can be found in [24].

A. Computational Geometry and Boundary Conditions

The Busemann inlet utilized in these simulations is the same as described in the experimental work of Schram et al. [10]. The inlet was designed for Mach 5.5 operation with a frontal diameter of 44 mm with a contraction ratio of 3:1. The leading edge has a uniform 2 mm radius. The total length of the inlet is 277 mm and a 51 mm constant area isolator section is added onto the end of the inlet. The geometry is shown in Figure 1. The inlet is placed within a rectangular test section, which has a length of 650 mm and a 150mm x 150mm cross-sectional area. The inlet is run at three separate angle of attacks, which are 0° and $\pm 5^\circ$. The inlet was run at a sub-design Mach number of $Ma = 3.7$ with a freestream pressure and temperature of $P_\infty = 10.9$ kPa and $T_\infty = 79.7$ K, respectively.

The computational domain consisted of the entire wind tunnel test section. In order to reduce cell count, cell clustering is primarily focused on the inlet internal surfaces, leading edge, and bleed slot. The centerplane cross-section of the mesh is depicted in Figure 1. The resulting mesh size was approximately 9 million tetrahedral elements and similar sizing was used for all angle of attack conditions. All simulations are presented using third order accurate DG($p=2$) elements as this was shown to provide reasonable results on coarse grids in previous supersonic flow simulations [25].

III. Experimental Validation

Comparisons are made between the simulated wall pressure and experimental pressure sensitive paint (PSP) measurements from Schram et al. to validate the simulation approach [10]. Figure 4a presents contours of the top wall pressure distribution for each angle of attack along the top wall of the inlet. These results show that wall pressure increases as the angle of attack rises and highlight how downstream shock-wall interactions lead to local pressure rises

along the wall. Centerline pressure profiles from the simulation and PSP measurements are shown in 4b. For $\alpha = -5^\circ$ and $\alpha = 0^\circ$ conditions, there is reasonable agreement between the simulations and experimental PSP measurements. However, at the $\alpha = +5^\circ$ condition, the simulation overpredicts the wall pressure. This behavior is consistent with RANS simulation results presented by San Martin et al. on the same configuration [26].

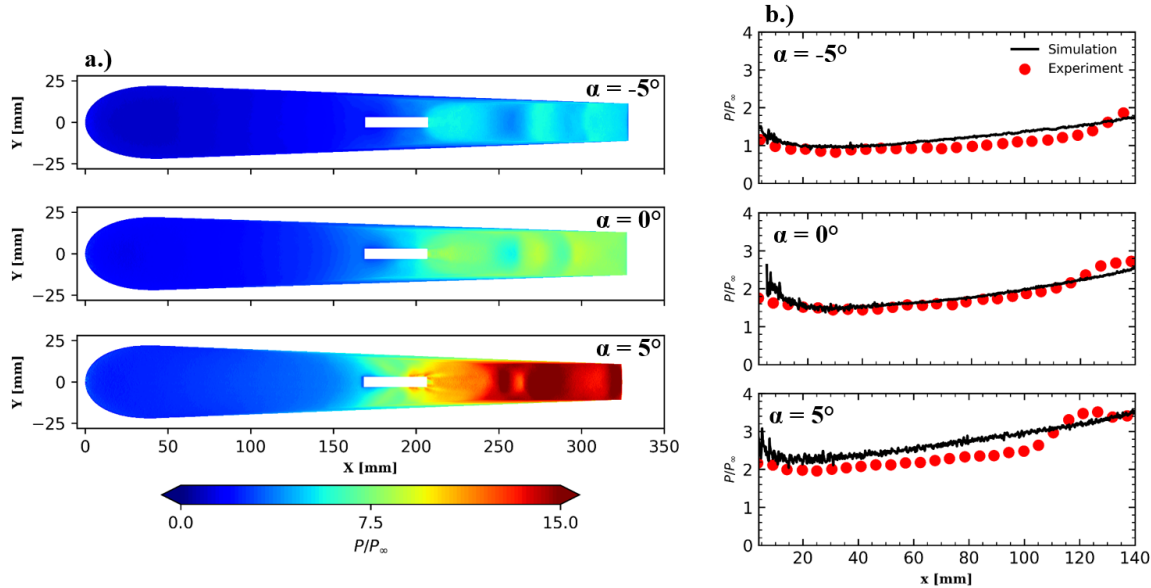


Fig. 2 Time averaged a.) top wall pressure distributions at $\alpha = \pm 5$ and 0 and b.) centerline pressure comparisons with experimental measurements

Additionally, schlieren images are available to identify the leading edge shock structures caused by the inlet for condition. Figure 3 presents contours of the density gradient to compare with experimental schlieren imaging. Examining the leading shock for each configuration shows that under negative angle of attack conditions, the shock resides outside the body of the inlet and is significantly detached from the cowl region. At level flight, the shock begins to move inside the inlet. At these conditions, the simulations show reasonable agreement with the shock locations as presented in the experimental works by Schram et al[27]. However, at the $\alpha = 5^\circ$, the simulated shock moves further into the inlet body and moves closer to the cowl. In contrast, the experimental results indicated that the leading shock remained further from the cowl. This deviation could also account for the pressure discrepancy noted in Figure 2 and should be further investigated.

IV. Results

The flowfield of the inlet is first examined for each angle of attack. Time-averaged contours of Mach number and pressure are presented in Figure 4. A primary difference in the flowfield is that as the angle of attack increases from -5° to $+5^\circ$, the pressure increases from approximately 4 P/P_∞ to 15 P/P_∞ . This is due to the change in flow direction, which causes the air to impinge more directly on the top wall. The Mach number contours reflect a similar trend, in which the inlet Mach number decreases by approximately 50% as the angle of attack is increased to its maximum value as a result of the pressure rise.

To further evaluate the inlet's performance, YZ plane contours are extracted downstream of the bleed slot to quantify flow characteristics, uniformity, and mass flow rate. Instantaneous contours of Mach number are presented in Figure 5. Table 1 summarizes the captured mass flow and mass flow loss for each angle of attack. The total captured mass flow is calculated at a plane slightly downstream of the inlet cowl, while the lost mass flow to bleed is determined from the planes extracted in Figure 5. It is shown that the mass capture rises as the angle of attack increases. However, there is minimal difference in mass flow loss to bleed between $\alpha = 0^\circ$ and $\alpha = +5^\circ$ angle of attack cases, likely a result of similar boundary layer behavior depicted in Figure 4.

The instantaneous contours in Figure 5 reveal distinct flow field characteristics within the inlet cross-section. At the

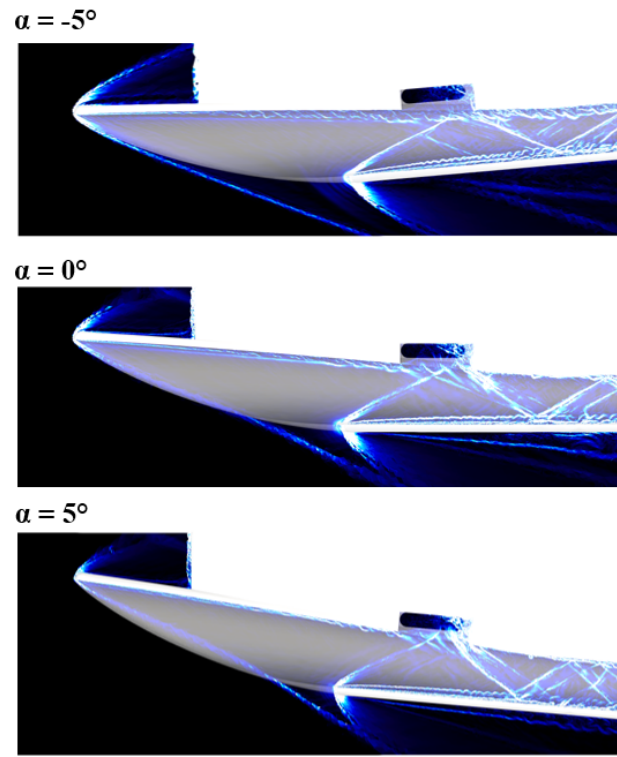


Fig. 3 Instantaneous density gradient contours for each angle of attack

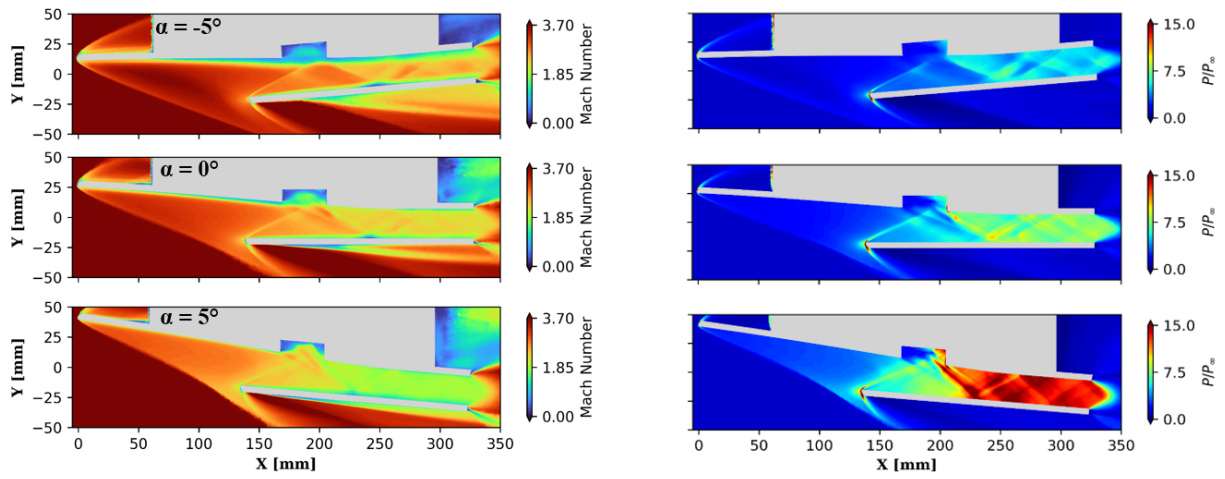


Fig. 4 Time averaged centerplane contours of Mach number and pressure at $\alpha = \pm 5$ and 0

α	-5°	0°	$+5^\circ$
$\dot{m}$ (kg/s)	0.0340	0.0455	0.0588
$\dot{m}_{loss}$ (%)	3.46	4.61	4.75

Table 1 Mass flow rate captured and mass flow rate lost due at each operating condition

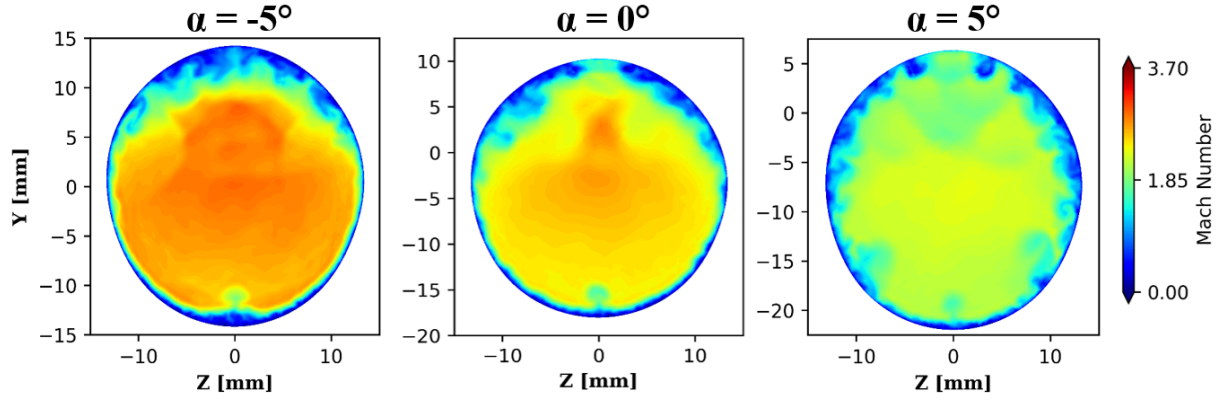


Fig. 5 Instantaneous contours of Mach Number for $\alpha = \pm 5$ and 0

+5° angle of attack condition, the cross-section exhibits the most widespread turbulent structures which are represented by the low Mach roll-up regions along the inlet walls. In contrast, for $\alpha = -5^\circ$ and $\alpha = 0^\circ$ cases, these regions are primarily confined to the top wall, diminishing significantly near the side and bottom wall regions. An additional key difference is that when the inlet is pitched down (-5°), there is no significant disruption of the incoming boundary layer from the bleed slot. However, in both the 0° and $+5^\circ$ cases, a noticeable breakup of the boundary layer structure occurs in this region.

To further quantify the effect of angle of attack on pressure and Mach number through the inlet, average pressure and Mach number values are calculated at different axial locations along the inlet length and presented in Figure 6. It is seen that the pressure increases 80% when the inlet is pitched up ($\alpha = +5^\circ$) when compared to its nominal ($\alpha = 0^\circ$) configuration. Similarly, due to this more significant pressure rise, the Mach number of the flow near the inlet drops by approximately 20%. When the inlet is pitched down ($\alpha = -5^\circ$) the Mach number at the exit of the inlet increases by approximately 10% and the pressure decreases by 30%. These statistics indicate that a significantly different flowfield will be entering the isolator and combustor which significantly effect the overall performance of the vehicle.

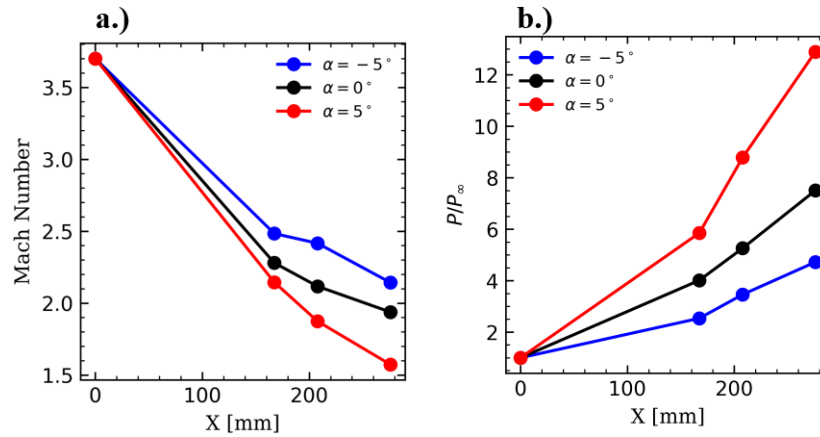


Fig. 6 Average a.) Mach number and b.) Pressure extracted at different YZ planes

To further quantify the inlet's performance at different angles of attack, contours of total pressure recovery and turbulent kinetic energy (TKE) are presented in Figure 7 and Figure 8, respectively. The total pressure recovery contours are normalized by the freestream value to quantify the losses through the inlet. Analysis is conducted at three axial locations: 167 mm (prior to the bleed slot), 208 mm (slightly downstream of the bleed slot), and 276 mm (near the inlet exit). These locations are selected to capture the inlet capture and inlet outflow characteristics. At the initial plane ($x =$

167 mm), a key observation across all conditions is the total pressure losses within the boundary layer along the top half of the inlet. For $\alpha = -5^\circ$, the boundary layer is thickest, resulting in the most severe total pressure recovery reduction in the center of the top wall. During level operation ($\alpha = 0^\circ$), the region of low total pressure recovery distributes more evenly, wrapping around to the inlet side walls. When the inlet is pitched up ($\alpha = +5^\circ$), there are fewer losses along the top wall; however, two distinct lobes of reduced recovery emerge along the bottom wall, a feature absent in the other configurations. These general flow features largely persist at the $x = 208$ mm location. A key difference at this plane is how the bleed slot breaks up the total pressure losses along the top wall in both $\alpha = 0^\circ$ and $+5^\circ$ configurations. For the pitched down ($\alpha = -5^\circ$) configuration, the boundary layer remains attached, and its associated losses remain relatively constant across all examined planes. At the final plane ($x = 276$ mm) near the inlet exit, each configuration exhibits a unique spatial distribution of total pressure losses. For $\alpha = -5^\circ$, the region of highest total pressure recovery forms a prominent semicircle encompassing the bottom half of the inlet cross-section. At $\alpha = 0^\circ$, optimal recovery is observed in a pear-shaped region extending into the top half of the inlet. Lastly, in the pitched up configuration ($\alpha = +5^\circ$), the highest recovery is concentrated in a narrowband along the very center of the inlet.

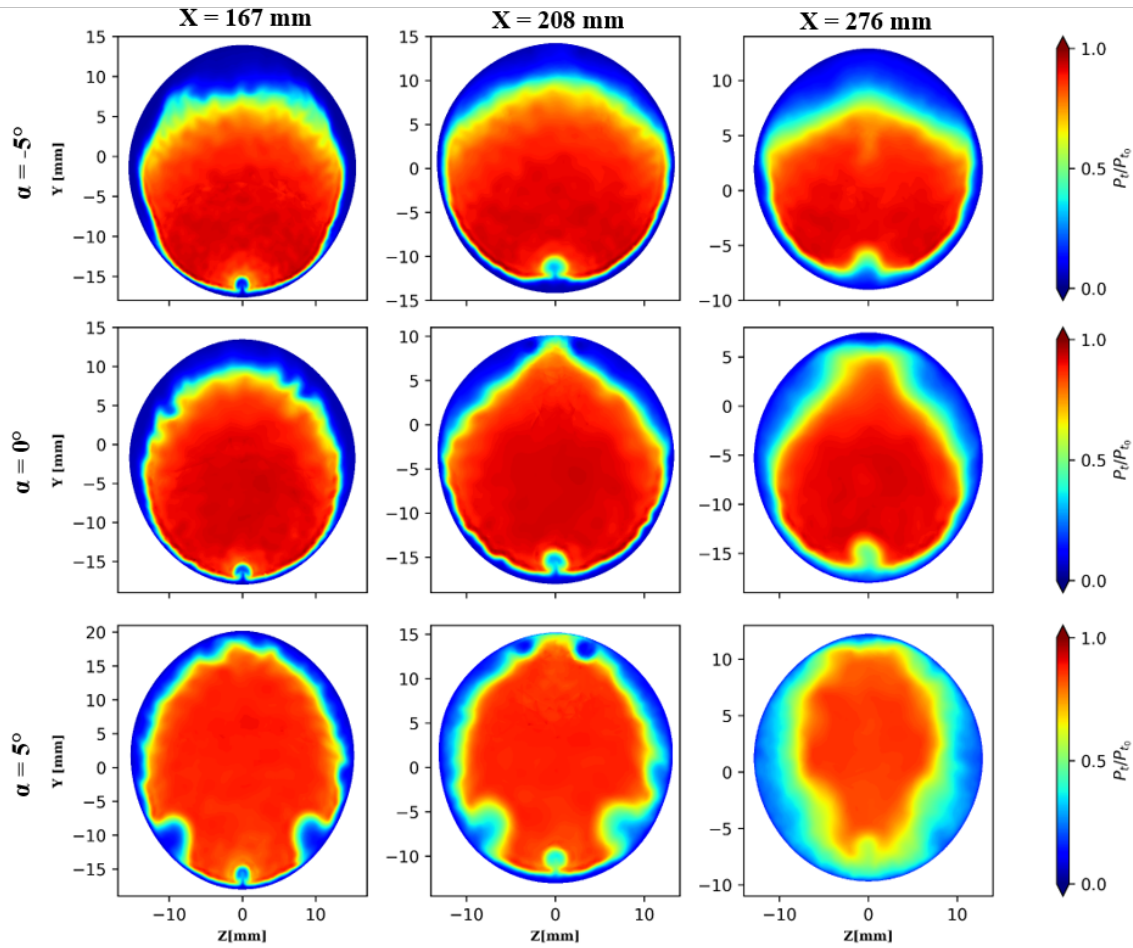


Fig. 7 Average total pressure recovery at each angle of attack at different locations along the inlet length

Further analysis of inlet flow uniformity includes examining the turbulent kinetic energy (TKE) distributions at the same axial locations. In general, the TKE distributions closely mirror the total pressure recovery contours, as high TKE production is directly correlated to regions of low total pressure recovery. For all angles of attack, TKE is predominantly concentrated within the boundary layer region. At $\alpha = -5^\circ$ and 0° the TKE is primarily located along the upper half of the inlet, with the $\alpha = 0^\circ$ distribution showing a clear bias towards the top sidewalls. However, when the inlet is pitched

up, the TKE has a much more even distribution across the entire inlet cross-section, particularly as it approaches the inlet exit. This TKE production creates conditions that directly impact shock-boundary layer interactions within the isolator and fuel injection and combustion characteristics downstream in the combustor.

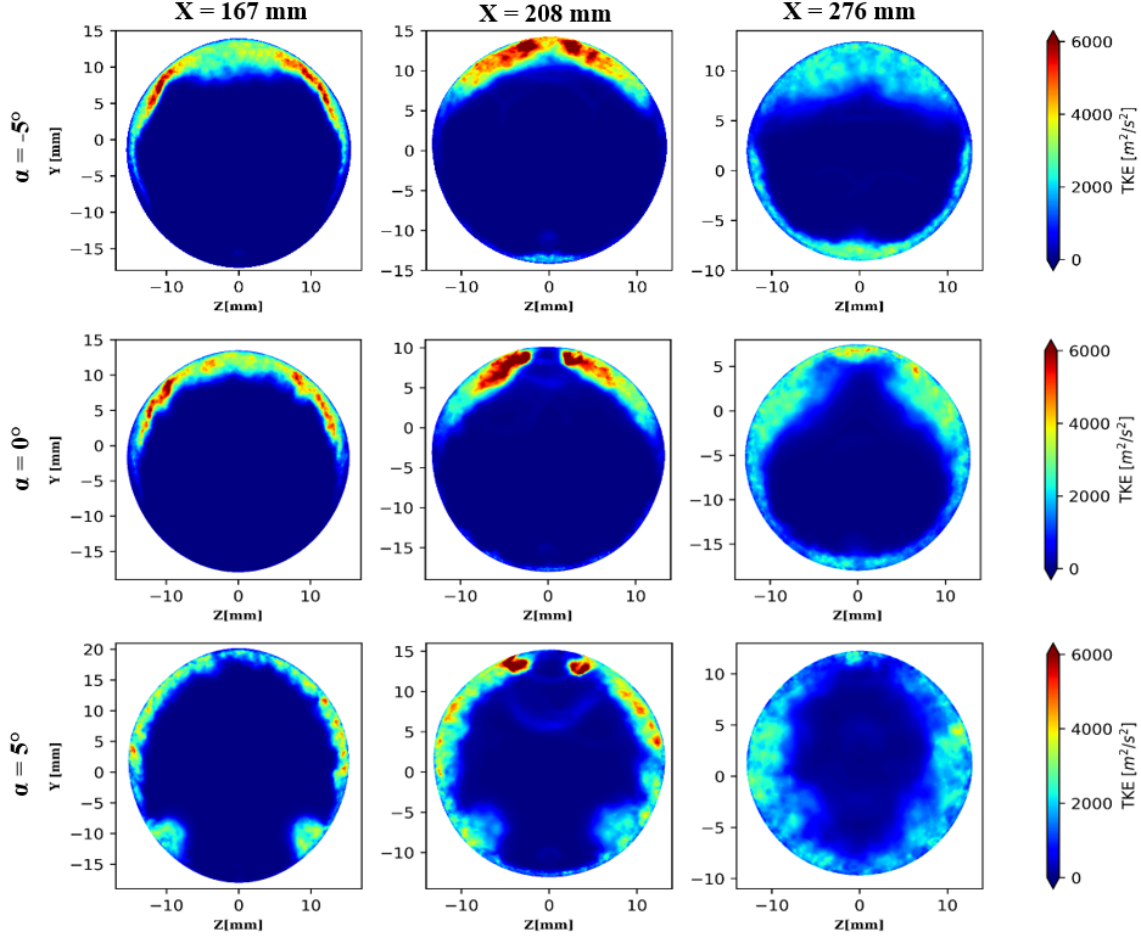


Fig. 8 Average turbulent kinetic energy at each angle of attack at different locations along the inlet length

V. Conclusions

Implicit Large Eddy Simulation (LES) with a high-order discontinuous Galerkin method on unstructured tetrahedral meshes was utilized to simulate an underspeed Busemann inlet across varying angles of attack. A $p = 2$ polynomial approximation, corresponding to nominally third-order accuracy, was employed for all simulations. Validation against experimental pressure and schlieren data showed good overall agreement, though some discrepancies in shock location and wall pressure were observed for the $\alpha = 5^\circ$ case. The simulations revealed that angle of attack significantly influences the inlet's internal flowfield characteristics and performance. Pitching the inlet up resulted in a substantial increase in pressure, while pitching it down caused a decrease. Notably, at negative angles of attack, the boundary layer along the inlet's top wall remained thick and attached, even across the bleed slot.

Cross-sectional analysis along the axial direction further revealed distinct flow structures for each angle of attack, including the formation of unique lobe structures along the bottom wall exclusively in the pitched-up configuration. Furthermore, total pressure recovery and turbulent kinetic energy (TKE) distributions were found to be closely correlated. Changes in angle of attack directly influenced the spatial distribution of TKE production and areas of high total pressure recovery. Specifically, when the inlet was pitched up, a significant shift in TKE generation contributed to an overall

lower total pressure recovery across the inlet.

These results underscore the critical impact of inlet orientation on the flow quality delivered to downstream engine components, such as the isolator and combustor. Future work should expand the simulated angle of attack range to thoroughly assess operability limits and uncover further flowfield characteristics. Furthermore, simulations should be performed examining inlet unstart and influence both Mach number and angle of attack have on operability limits. Additionally, enhancing solution accuracy for pitched-up configurations through more extensive mesh refinement and the application of advanced meshing strategies, such as adaptive mesh refinement, is recommended.

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